

Information Field Theory as Artificial Intelligence

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Abstract

Information field theory (IFT), the information theory for fields, is a mathematical framework for signal reconstruction and non-parametric inverse problems. Here, fields denote physical quantities that change continuously as a function of space (and time) and *information theory* refers to Bayesian probabilistic logic equipped with the associated entropic information measures. Reconstructing a signal with IFT is a computational problem similar to training a *generative neural network* (GNN). In this paper, the inference in IFT is reformulated in terms of GNN training and the cross-fertilization of numerical variational inference methods used in IFT and machine learning are discussed. The discussion suggests that IFT inference can be regarded as a specific form of *artificial intelligence*. In contrast to classical neural networks, IFT based GNNs can operate without pre-training thanks to incorporating expert knowledge into their architecture.

1 Motivation

Determining the concrete configuration of a field from measurement data is an ill-posed inverse problem as physical fields have an infinite number of degrees of freedom (DoF), whereas data sets are always finite in size. Thus, the data provides a finite number of constraints for only a subset of the infinitely many DoF of a field. In order to infer a field, the remaining of its DoF need therefore to be constrained via prior information. Fortunately, physics provides us with such prior information on fields. This information might either be precise, like $\nabla \cdot B = 0$ in electrodynamics, or more phenomenological, in the sense that a field shaped by a certain process can often be characterized by its n -point correlation functions. Having knowledge on such correlations can be sufficient to regularize the otherwise ill-posed field inference problem from finite and noisy data such that meaningful statements about the field can be made.

As a formalism for this, *Information field theory* (IFT) was introduced [1, 2] following and extending earlier lines of work [e.g. 3, 4]. IFT is information theory for fields. It investigates the space of possible field configurations and constructs probability densities over those spaces in order to permit Bayesian field inference from data. It has been applied successfully to a number of problems in astrophysics [5–18], particle physics [19–21], and elsewhere [22–26]. Here, the relation of IFT with *artificial intelligence* (AI) methods is outlined, in particular with *generative neural networks* (GNNs). The presented line of arguments summarizes a number of recent works [27–32].

IFT is briefly introduced in Sec. 2 in its most modern incarnation in terms of standardized, generative models.

These are shown to be equivalent to GNNs in Sec. 3. The structural similarity of IFT inference and GNN training problems allows for a common set of *variational inference* (VI) methods, as discussed in Sec. 4. Sec. 5 concludes the argumentation on why IFT can be seen as a specific form of AI.

2 Information Field Theory

IFT allows to deduct fields from data in a probabilistic way. In order to be able to apply probability theory onto the space of field configurations, a measure in this space is needed. Although no canonical mathematical measure on function spaces exists, for IFT applications the usage of Gaussian process measures [33], which are mathematically well defined [34, 35], is usually fully sufficient. Gaussian processes can also be argued to be a natural starting point for reasoning on fields with known finite first and second order moments, as we will discuss now.

To be specific, let $\varphi : \Omega \rightarrow \mathbb{R}$ be a scalar field over some domain $\Omega \subset \mathbb{R}^u$ and our prior knowledge on φ be the first and second moments of the field, e.g.

$$\langle \varphi^x \rangle_{(\varphi)} = \bar{\varphi}^x \text{ and} \quad (1)$$

$$\langle (\varphi - \bar{\varphi})^x (\varphi - \bar{\varphi})^y \rangle_{(\varphi)} = \Phi^{xy} \text{ for all } x, y \in \Omega, \quad (2)$$

with $\varphi^x := \varphi(x)$ denoting a field value and $\langle f(\varphi) \rangle_{(\varphi)} := \int \mathcal{D}\varphi \mathcal{P}(\varphi) f(\varphi)$ a prior expectation value for some function f of the field. If only the first and second field moments are given as prior information, it follows from the maximum entropy principle that the least informative probability distribution function encoding this information is a Gaussian

with these moments. Thus using this Gaussian

$$\begin{aligned}\mathcal{P}(\varphi|I) &\equiv \mathcal{G}(\varphi - \bar{\varphi}, \Phi) \\ &:= \frac{1}{\sqrt{|2\pi\Phi|}} \exp\left(-\frac{1}{2}(\varphi - \bar{\varphi})^\dagger \Phi^{-1}(\varphi - \bar{\varphi})\right)\end{aligned}\quad (3)$$

as a prior with background information $I = (\langle\varphi\rangle_{(\varphi)} = \bar{\varphi}, \langle(\varphi - \bar{\varphi})(\varphi - \bar{\varphi})^\dagger\rangle_{(\varphi)} = \Phi)$ is a conservative choice, as it makes the least additional assumptions about the field except for the moments specified in I .

In many applications, however, the field of interest, the signal s , is not a Gaussian field, but may be related to such via a non-linear transformation. For example in astronomical applications of IFT, the sky brightness field s is the quantity of interest, which is strictly positive and therefore can not be a Gaussian field. However, the logarithm of a brightness can be positive and negative and may therefore be modeled as a Gaussian process. In such a case one could assign e.g. $s^x(\varphi) = s_0 \exp(\varphi^x)$ as a model for a diffuse (spatially correlated) sky emission component, with s_0 a reference brightness, chosen such that for example $\langle\varphi^x\rangle_{(\varphi)} = 0$ holds.

Having now a field prior, Bayesian reasoning on the field φ , and therefore on the signal of interest $s = s(\varphi)$, based on some data d and its likelihood $\mathcal{P}(d|\varphi, I)$ is possible. The field posterior

$$\mathcal{P}(\varphi|d, I) = \frac{\mathcal{P}(d|\varphi, I)\mathcal{P}(\varphi|I)}{\mathcal{P}(d|I)} \quad (4)$$

is defined as well as the prior and permits to answer questions about the field, like its most probable configuration $\varphi_{\text{MAP}} = \text{argmax}_{\varphi} \mathcal{P}(\varphi|d, I)$ (MAP = maximum a posteriori), its posterior mean $m = \langle\varphi\rangle_{(\varphi|d, I)}$, or its posterior uncertainty dispersion $D = \langle(\varphi - m)(\varphi - m)^\dagger\rangle_{(\varphi|d, I)}$. IFT exploits the formalism of quantum and statistical field theory to calculate such posterior expectation values [1, 28, 36–38]. These formal calculations, however, should not be the focus here. Instead, it should be the formulation of IFT inference problems in terms of generative models as these can be interpreted as GNNs.

For this purpose, the likelihood is expressed in terms of a measurement equation

$$d = R(\varphi) + n, \text{ with} \quad (5)$$

$$R(\varphi) := \langle d \rangle_{(d|\varphi)}, \quad (6)$$

$$n := d - R(\varphi), \text{ and} \quad (7)$$

$$\mathcal{P}(d|\varphi, I) \equiv \mathcal{P}(n = d - R(\varphi)|\varphi), \quad (8)$$

which is always possible if the data can be embedded into a vector space and the data expectation value defining the response function R exists. Here and in the following, we dropped denoting background information I in probabilities. This rewriting of the likelihood in terms of a mean instrument response $d' = R(\varphi)$ to the field φ and a noise process $\mathcal{P}(n|\varphi)$, which summarizes the fluctuations around

that mean d' , allows to regard the data as the result of a noisy generative process that maps field values φ and associated noise realizations n onto data d according to Eq. 5.

In case the instrument response and noise processes are provided for the signal s instead of the Gaussian field φ as $R'(s) := \langle d \rangle_{(d|s)}$ and $\mathcal{P}(n|s)$, their respective pull backs $R(\varphi) := \langle d \rangle_{(d|s(\varphi))} = R'(s(\varphi))$ and $\mathcal{P}(n|\varphi) := \mathcal{P}(n|s(\varphi))$ provide the necessary response and noise statistics w.r.t. the field φ .

All this provides a generative model for the signal s and data d via $\varphi \leftarrow \mathcal{G}(\varphi, \Phi)$, $s = s(\varphi)$, $n \leftarrow \mathcal{P}(n|s)$, and $d = R'(s) + n$, which should now be standardized.

Standardization of a random variable φ refers to finding a mapping from a standard normal distributed random variable $\xi \leftarrow \mathcal{G}(\xi, \mathbb{1})$ to φ that reproduces the statistics of $\mathcal{P}(\varphi|I)$. For a Gaussian field φ , this is just a mapping of the form

$$\varphi(\xi) := \Phi^{\frac{1}{2}}\xi, \quad (9)$$

where $\Phi^{\frac{1}{2}}$ refers to a square root of Φ , which always exists for a covariance matrix that is positive definite. For the large class of band diagonal and therefore translational invariant covariance matrices Φ , which are very relevant for applications as we argue below, the square root of Φ can be explicitly constructed.

In many signal inference problems, no spatial location is singled out a priori, before the measurement. This means that the field covariance between two locations only depends on the distance between these positions, but not on their absolute positions. Thus, $\Phi^{xy} = C_\varphi(x - y)$. As a consequence of the Wiener-Khintchine theorem, such a translational invariant field covariance becomes diagonal in harmonic space,

$$\tilde{\Phi}^{kq} = \mathcal{F}_x^k \Phi^{xy} (\mathcal{F}_y^\dagger)^q = (2\pi)^u \delta(k - q) P_\varphi(k) = \widehat{P}_\varphi^{kq}. \quad (10)$$

Here and in the following, $\mathcal{F}$ denotes a harmonic transform (a u -dimensional Fourier transform $\mathcal{F}_x^k = \exp(ik \cdot x)$ in case of an Euclidean space, as we assume in the following), $\dagger$ the adjointed (complex conjugate and transposed of a matrix or vector), $P_\varphi(k) := \mathcal{F}_x^k C_\varphi^{xx}$ is the so called power spectrum of φ , the Einstein convention for repeated indices is used, as in $\tilde{\varphi}^k := \mathcal{F}_x^k \varphi^x \equiv \int dx^u \exp(ik \cdot x) \varphi(x)$, and $\hat{\phi} = \text{diag}(\phi)$ denotes a diagonal operator in the space of the field ϕ with the values of ϕ on the diagonal.

Thanks to this diagonal representation of the field covariance in harmonic space, an explicit standardization of the field is given via

$$\xi \leftarrow \mathcal{G}(\xi, \mathbb{1}), \quad (11)$$

$$\varphi = \mathcal{F}^{-1} A \xi, \text{ and} \quad (12)$$

$$A_\varphi = \widehat{P_\varphi^{1/2}}, \quad (13)$$

where the latter is an amplitude operator that is diagonal in harmonic space and that imprints the right amplitudes

onto the Fourier modes of φ . This can be seen via a direct calculation,

$$\begin{aligned}\langle \varphi(\xi) \varphi(\xi)^\dagger \rangle_{(\xi)} &= \mathcal{F}^{-1} A_\varphi \langle \xi \xi \rangle_{(\xi)} A_\varphi^\dagger \mathcal{F}^{-1\dagger} \\ &= \mathcal{F}^{-1} A_\varphi \mathbb{1} A_\varphi^\dagger \mathcal{F}^{-1\dagger} \\ &= \mathcal{F}^{-1} \tilde{\Phi} \mathcal{F}^{-1\dagger} = \Phi \\ &= \langle \varphi \varphi^\dagger \rangle_{(\varphi)}.\end{aligned}\quad (14)$$

In case no direction of the space is singled out a priori, the two-point correlation function and the power spectrum of φ become isotropic, $\Phi^{xy} = C_\varphi(|x - y|)$ and $\tilde{\Phi}^{kq} = (2\pi)^u \delta(k - q) P_\varphi(|k|)$, respectively. In this case only a one dimensional power spectrum needs to be known. Such power spectra are often smooth functions on a double logarithmic scale in Fourier space, since any sharp feature in them would correspond to a (quasi-)periodic pattern in position space, which would be very unnatural for most signals. Thus, introducing the logarithmic Fourier space scale variable $\kappa(k) := \ln k/k_0$ w.r.t. some reference scale k_0 , we expect

$$\psi(\kappa) := \ln(P_\varphi(k_0 e^\kappa)/P_0) \quad (15)$$

to be a field itself, in the sense that it is sufficiently smooth. Here, P_0 is a pivot scale for the power spectrum.

Often, the power spectrum as parameterized through ψ is not known a priori for a field φ , but statistical homogeneity, isotropy, and the absence of long range quasi-periodic signal variations make a Gaussian field prior for ψ plausible, $\mathcal{P}(\psi) = \mathcal{G}(\psi - \bar{\psi}, \Psi)$. This log-log-power spectrum may exhibit fluctuations $\chi := \psi - \bar{\psi}$ around a non-zero mean $\bar{\psi}(\kappa)$. The latter might e.g. encode a preference for falling spectra and therefore for a spatially smooth field φ . In this case just another layer for χ of a standardized generative model has to be added,

$$\eta \leftarrow \mathcal{G}(\eta, \mathbb{1}) \quad (16)$$

$$\chi(\eta) := A_\psi \eta, \text{ with} \quad (17)$$

$$A_\psi A_\psi^\dagger := \Psi, \text{ and} \quad (18)$$

$$\psi(\eta) := \bar{\psi} + \chi(\eta). \quad (19)$$

Again, a prior for a field, here the only one dimensional $\chi(\kappa)$, is needed. A detailed description of how this can be provided in an efficient way is given by [15]. This reference also provides a generative model for the case that the signal domain Ω is a product of sub-spaces, like position space and an energy spectrum coordinate, each requiring a different correlation structure, and the total correlation being a direct product of those. Assuming a direct product for the correlation structures might be possible for many field inference problems [as, e.g. in 39, 15].

Dynamical systems, however, typically exhibit correlation structures that are not direct products of the spatial and temporal sub-spaces. Here, the full spatial and temporal Fourier power spectrum $P_\varphi(k, \omega)$, with ω being the temporal frequency, encodes the full dynamics of linear, autonomous system. An inference of this highly structured

spectrum from data can be used to learn the system dynamics from system measurements [26, 40]. It just requires a more complex spectral prior than discussed here.

To summarize, the field inference problems of IFT can often be stated in terms of a standardized, generative model for the signal and the data. For the illustrative case outlined above, where the probabilistic model is given by

$$\mathcal{P}(d, \varphi, \psi) = \mathcal{P}(d|\varphi) \mathcal{P}(\varphi|\psi) \mathcal{P}(\psi), \quad (20)$$

$$\mathcal{P}(\psi) = \mathcal{G}(\psi - \bar{\psi}, \Psi), \quad (21)$$

$$\mathcal{P}(\varphi|\psi) = \mathcal{G}(\varphi, \Phi(\psi)), \quad (22)$$

$$\Phi(\psi) = \mathcal{F}^{-1} \widehat{P}_\varphi \mathcal{F}^{-1\dagger}, \quad (23)$$

$$P_\varphi(k) = P_0 \exp(\psi(k_0 e^k)), \text{ and} \quad (24)$$

$$\mathcal{P}(d|\varphi) = \mathcal{P}(d|s(\varphi)), \quad (25)$$

the corresponding standardized generative model is

$$\zeta := (\xi, \eta) \leftarrow \mathcal{G}(\zeta, \mathbb{1}), \quad (26)$$

$$\psi(\eta) := \bar{\psi} + A_\psi \eta, \quad (27)$$

$$P_\varphi(k) := P_0 \exp(\psi(k_0 \exp(|k|))), \quad (28)$$

$$\varphi(\xi, \psi) := \mathcal{F}^{-1} \widehat{P}_\varphi^{1/2} \xi, \quad (29)$$

$$s(\varphi) := s_0 \exp(\varphi), \quad (30)$$

$$n \leftarrow \mathcal{P}(n|s), \text{ and} \quad (31)$$

$$d = R'(s) + n. \quad (32)$$

For the noiseless data $d' = R'(s)$ the generative model reads

$$\begin{aligned}d'(\zeta) &:= R'(s(\varphi(\xi, \psi(\eta)))) \\ &= (R' \circ s \circ \varphi \circ f)(\zeta), \text{ with}\end{aligned} \quad (33)$$

$$f(\zeta) := (\xi, \psi(\eta)). \quad (34)$$

This way, the full model complexity as given by Eqs. 20-25 has been transferred into an effective response function $d' = R' \circ s \circ \varphi \circ f$. For this latent variable vector, the prior is simply $\mathcal{P}(\zeta) = \mathcal{G}(\zeta, \mathbb{1})$, whereas the likelihood $\mathcal{P}(d|\zeta) = \mathcal{P}(n = d - d'(\zeta)|\zeta)$ has absorbed the full model complexity. This so called reparametrization trick [41] was introduced to IFT by [29] to simplify numerical VI.

What is essential at this point is to realize that this generative model consists of a latent space white noise process $\mathcal{P}(\zeta) = \mathcal{G}(\zeta, \mathbb{1})$ that generates an input vector ζ and a sequence of non-local linear and local non-linear operations that is applied to it. The Fourier transform $\mathcal{F}^{-1}$ and A_ψ are examples of non-local linear operations within the model. Among the non-linear operations are the exponential functions and the application of the ψ -dependent amplitude operator $A_\varphi(\psi)$ to the latent space excitations ξ , as there the two components of $\zeta = (\xi, \eta)$ are multiplied together. Furthermore, the instrument response $R'(s)$ might also be decomposed into sequences of non-local linear and local non-linear operations, as physical processes in measurement devices can often be cast into the propagation of a quantity (an operation that is linear in the quantity) and the local interactions of the quantity (an operation non-linear in it), respectively.

3 Artificial Intelligence

AI is a vast field. Here, the focus should be on GNNs. These are neural networks that transform a latent space variable $\xi \leftarrow \mathcal{G}(\xi, \mathbb{1})$ into a signal or data realization, $s = s(\xi)$ or $d' = d'(\xi)$. A neural network is a function $g(\xi)$ that can be decomposed in terms of a number n layer processing functions g_i with

$$g = g_n \circ g_{n-1} \circ \dots \circ g_1. \quad (35)$$

Any of the layer processing functions $g_i : \xi_i \mapsto \xi_{i+1}$ with $\xi_1 \equiv \xi$ consists typically of a non-local, affine linear transformation $l_i(\xi_i) := L_i \xi_i + b_i$ of the input vector ξ_i of layer i followed by a local, point wise application of non-linear, so-called activation functions $\sigma_i : \mathbb{R} \rightarrow \mathbb{R}$. Thus, the output vector ξ_{i+1} of layer i is

$$\xi_{i+1} = (\sigma_i \circ l_i)(\xi_i), \quad (36)$$

where σ_i acts component wise. The set $\eta = (L_i, b_i)_{i=1}^n$ of all coefficients of the l_i s (the matrix elements of the L_i matrices, and the compnents of the b_i vectors) determines the function the network represents. Putting the input values and network coefficients into a single vector $\zeta := (\xi, \eta)$ a GNN can be regarded as a function of both, latent variables ξ and network parameters η , $d'(\zeta) = g(\zeta; \eta)$.

From this perspective, a standardized, generative model $d'(\zeta)$ in IFT is structurally a GNN, as both consists of sequences of local non-linear and non-local linear operations on their input vector $\zeta = (\xi, \eta)$. The way IFT-based models and GNNs are usually used differs a bit. Both can be used to generate synthetic samples of outputs by processing random latent space vectors $\xi \leftarrow \mathcal{G}(\xi, \mathbb{1})$. However, typically an IFT model $d'(\zeta)$ is applied to infer all latent space variables in ζ from data d . From the latent variables, the signal of interest can always be recovered via $s(\zeta)$.

For this inference the so-called information Hamiltonian

$$\begin{aligned} \mathcal{H}(d, \zeta|I) &= -\ln \mathcal{P}(d, \zeta|I) = -\ln \mathcal{P}(d|\zeta, I) - \ln \mathcal{P}(\zeta|I) \\ &= \mathcal{H}(n = d - d'(\zeta)|\zeta, I) + \frac{1}{2}\zeta^\dagger \zeta + \text{const} \end{aligned} \quad (37)$$

is investigated w.r.t. ζ , where $\mathcal{H}(a|b) := -\ln \mathcal{P}(a|b)$. This is done, for example, by minimizing $\mathcal{H}$ to obtain a MAP estimator for ζ or – as discussed in the next section – via VI. In case of a constant, signal independent Gaussian white noise statistics, the information Hamiltonian becomes

$$\mathcal{H}(d, \zeta|I) = \frac{|d - d'(\zeta)|^2}{2\sigma_n^2} + \frac{1}{2}\zeta^\dagger \zeta + \text{const.} \quad (38)$$

The training of an usual GNN is done with a training data set $\tilde{d} = (d_i)_i$ to which a corresponding latent space vector set $\tilde{\zeta} = (\zeta_i)_i$ and common network parameters η

needs to be found. For this a loss function of the form

$$\tilde{\mathcal{H}}(\tilde{d}, \tilde{\zeta}, \eta) = \sum_i \tilde{\mathcal{H}}(d_i|\xi_i, \eta) + \tilde{\mathcal{H}}(\tilde{\zeta}|\eta) + \tilde{\mathcal{H}}(\eta), \quad (39)$$

$$\tilde{\mathcal{H}}(\tilde{\zeta}|\eta) = \frac{1}{2} \sum_i \xi_i^\dagger \xi_i + \text{const}, \text{ and} \quad (40)$$

$$\tilde{\mathcal{H}}(d_i|\xi_i, \eta) = \frac{1}{2} \frac{|d_i - d'(\xi_i, \eta)|^2}{2\sigma_n^2} + \text{const} \quad (41)$$

might be minimized. Here, a typical GNN data loss function $\tilde{\mathcal{H}}(d_i|\xi_i, \eta)$ as used for the decoder part of an *autoencoder* (AE) [42] was assumed. In an *generative adversarial network* (GAN) [43], however, this data loss function is given in terms of the output of a discriminator network. The network parameter prior term $\tilde{\mathcal{H}}(\eta)$ might be chosen to be uninformative ($\tilde{\mathcal{H}}(\eta) = \text{const}$) or informative (e.g. $\tilde{\mathcal{H}}(\eta) = \frac{1}{2}\eta^\dagger \eta$ in case of a Gaussian prior on the parameters).

Anyhow, by comparison of Eqs. 39-41 with Eqs. 37 and 38 it should be apparent that the network loss functions can be structurally similar to the IFT information Hamiltonian. Both can have a probabilistic interpretation in terms of being negative log-probabilities, e.g.

$$\mathcal{P}(d, \xi, \eta) = e^{-\mathcal{H}(d, \xi, \eta)} \text{ and} \quad (42)$$

$$\mathcal{P}(\tilde{d}, \tilde{\zeta}, \eta) = e^{-\tilde{\mathcal{H}}(\tilde{d}, \tilde{\zeta}, \eta)}, \quad (43)$$

respectively. For this reason, we do not distinguish between an information Hamiltonian $\mathcal{H}$ and a network loss function $\tilde{\mathcal{H}}$ by writing $\mathcal{H}$ for both in the following.

The IFT-GNN can operate with solely a single data vector d due to the domain knowledge coded into their architecture, whereas usual GNNs require sets of data vectors $\tilde{d} = (d_i)_i$ to be trained. Recently, more IFT-like architectures for GNNs were proposed as well, which are also able to process data without training [44].

4 Variational Inference

So far, it has been assumed here that MAP estimators are used to determine network parameters ζ for both, IFT-based models as well as traditional GNNs. MAP estimators are known to be prone to over-fitting the data, as they are not probing the adjacent phase-space volumes of their solutions. VI methods perform better in that respect, while still being affordable in terms of computational costs for the high dimensional settings of IFT-based field inference and traditional GNN training. They were used in most recent IFT applications [12–18, 21, 45] and are prominently present in the name of *variational autoencoders* (VAEs) [41] that are build on VI.

In VI, the posterior $\mathcal{P}(\zeta|d)$ is approximated by a simpler probability distribution, in many applications by a Gaussian

$$\mathcal{Q}(\zeta|\theta, \Theta) = \mathcal{G}(\zeta - \theta, \Theta). \quad (44)$$

The Gaussian is chosen to minimize the variational Kullback-Leibler (KL) divergence

$$\begin{aligned} \text{KL}_\zeta((\theta, \Theta), d) &:= \mathcal{D}_{\text{KL}}(\mathcal{Q}||\mathcal{P}) \\ &= \int \mathcal{D}\zeta \mathcal{Q}(\zeta|\theta, \Theta) \ln \frac{\mathcal{Q}(\zeta|\theta, \Theta)}{\mathcal{P}(\zeta|d)} \quad (45) \end{aligned}$$

w.r.t. the parameters θ and Θ of $\mathcal{Q}$. This is slightly sub-optimal from an information theoretical point of view as this minimizes the amount of information introduced by going from $\mathcal{P}$ to $\mathcal{Q}$. The *expectation propagation* (EP) KL with reversed arguments $\mathcal{D}_{\text{KL}}(\mathcal{P}||\mathcal{Q})$ would be better, as it minimizes the information loss from approximating $\mathcal{P}$ with $\mathcal{Q}$ [46]. VI is known to underestimate the uncertainties, whereas EP conservatively overestimates them. However, calculating the EP solution for θ and Θ would require integrating over the posterior. If this would be feasible, any posterior quantity of interest could be calculated as well and there would be no need to approximate $\mathcal{P}(\zeta|d)$ in the first place. Estimating and minimizing the VI KL $\mathcal{D}_{\text{KL}}(\mathcal{Q}||\mathcal{P})$ is less demanding, as the integral over the simpler (Gaussian) distribution $\mathcal{Q}$ can very often be performed analytically, or by sample averaging using samples drawn from $\mathcal{Q}$.

Thus, the posterior mean θ and the posterior uncertainty covariance Θ become parameters to be determined. The vector θ has the dimension N_{dim} of the latent space, whereas the posterior uncertainty covariance Θ has $N_{\text{dim}}(N_{\text{dim}} - 1)/2 = \mathcal{O}(N_{\text{dim}}^2)$ independent DoF. For small problems, these might be solved for, however, for large problems with millions of DoF these can even not be stored in a computer memory. To circumvent this, the *Automatic Differentiation Variational Inference* (ADVI) algorithm [47] often invokes the so called *mean field approximation* (MFA). This assumes a diagonal covariance $\Theta_{\text{MFA}} = \hat{\theta}' \equiv \text{diag}(\theta')$, with θ' being a latent space vector. Cross-correlations between parameters can not be represented by this, which is problematic in particular in combination with the tendency of VI to underestimate uncertainties.

In order to overcome this limitation of ADVI that limits its usage in IF-T contexts with their large number of DoF, the *Metric Gaussian Variational Inference* (MGVI) [30] algorithm approximates the posterior uncertainty of ζ with the help of the Fisher information matrix

$$M(\zeta) := \left\langle \frac{\partial \mathcal{H}(d|\zeta)}{\partial \zeta} \frac{\partial \mathcal{H}(d|\zeta)^\dagger}{\partial \zeta} \right\rangle_{(d|\zeta)}. \quad (46)$$

The starting point for obtaining the uncertainty covariance Θ used in MGVI is the Hessian of the log-posterior

$$\begin{aligned} \frac{\partial^2 \mathcal{H}(\zeta|d)}{\partial \zeta \partial \zeta^\dagger} &= \frac{\partial^2 \mathcal{H}(d, \zeta)}{\partial \zeta \partial \zeta^\dagger} - \underbrace{\frac{\partial^2 \mathcal{H}(d)}{\partial \zeta \partial \zeta^\dagger}}_{=0} \\ &= \frac{\partial^2 \mathcal{H}(d, \zeta)}{\partial \zeta \partial \zeta^\dagger} \end{aligned}$$

as a first guess for the approximate posterior precision matrix Θ^{-1} . Using this evaluated at the minimum ζ_{MAP} of the information Hamiltonian $\mathcal{H}(d, \zeta)$ would correspond to the Laplace approximation. However, neither is the MAP solution ideal, as discussed above, nor would this be a good approximation at many locations θ that differ from ζ_{MAP} . This is because positive definiteness of the Hessian is not guaranteed there, but it is an essential property of any correlation and precision matrix. For this reason, Θ^{-1} can not directly be approximated by this Hessian. However, the likelihood averaged Hessian is strictly positive definite, and can therefore serve as an approximate posterior precision matrix for any guessed posterior mean θ , as a short calculation shows:

$$\Theta^{-1}(\theta) \approx \left\langle \frac{\partial^2 \mathcal{H}(d, \zeta)}{\partial \zeta \partial \zeta^\dagger} \right\rangle_{(d|\zeta=\theta)} \quad (47)$$

$$= \left\langle \frac{\partial^2 \mathcal{H}(\zeta)}{\partial \zeta \partial \zeta^\dagger} + \frac{\partial^2 \mathcal{H}(d|\zeta)}{\partial \zeta \partial \zeta^\dagger} \right\rangle_{(d|\zeta=\theta)} \quad (48)$$

$$= \left\langle \mathbf{1} - \frac{\partial^2 \ln \mathcal{P}(d|\zeta)}{\partial \zeta \partial \zeta^\dagger} \right\rangle_{(d|\zeta=\theta)} \quad (49)$$

$$= \mathbf{1} - \left\langle \frac{1}{\mathcal{P}(d|\zeta)} \frac{\partial^2 \mathcal{P}(d|\zeta)}{\partial \zeta \partial \zeta^\dagger} \right\rangle_{(d|\zeta=\theta)} \quad (50)$$

$$+ \left\langle \frac{1}{\mathcal{P}^2(d|\zeta)} \frac{\partial \mathcal{P}(d|\zeta)}{\partial \zeta} \frac{\partial \mathcal{P}(d|\zeta)^\dagger}{\partial \zeta} \right\rangle_{(d|\zeta=\theta)} \quad (51)$$

$$= \mathbf{1} - \int dd \frac{\mathcal{P}(d|\zeta)}{\mathcal{P}(d|\zeta)} \frac{\partial^2 \mathcal{P}(d|\zeta)}{\partial \zeta \partial \zeta^\dagger} \quad (52)$$

$$+ \left\langle \frac{\partial \mathcal{H}(d|\zeta)}{\partial \zeta} \frac{\partial \mathcal{H}(d|\zeta)^\dagger}{\partial \zeta} \right\rangle_{(d|\zeta=\theta)} \quad (53)$$

$$= \mathbf{1} - \underbrace{\frac{\partial^2}{\partial \zeta \partial \zeta^\dagger} \int dd \mathcal{P}(d|\zeta)}_{=0} + M(\theta) \quad (54)$$

$$= \mathbf{1} + M(\theta) > 0. \quad (55)$$

The Fisher metric $M(\theta)$ is positive semidefinite as being an average over outer products ($v v^\dagger \geq 0$) of likelihood Hamiltonian gradient vectors $v = \partial \mathcal{H}(d|\zeta)/\partial \zeta$. Adding $\mathbf{1} > 0$ to the Fisher matrix turns the approximate precision matrix into a positive definite matrix $\Theta^{-1}(\theta) > 0$, of which the inverse $\Theta(\theta)$ exists for all θ , and which is positive definite as well.

This likelihood averaged Laplace approximation actually becomes exact in case of linear Gaussian measurement problems as is shown in the following. For such problems the measurement equation is of the form $d = R\zeta + n$, the noise statistics $\mathcal{P}(n|\zeta) = \mathcal{G}(n, N)$, and the standardized prior is $\mathcal{P}(\zeta) = \mathcal{G}(\zeta, \mathbf{1})$. The corresponding posterior is known to be a Gaussian

$$\mathcal{P}(\zeta|d) = \mathcal{G}(\zeta - m, D)$$

with mean m and covariance D given by the generalized

Wiener filter solution $m = D R^\dagger N^{-1} d$ and the Wiener covariance $D = (\mathbb{1} + R^\dagger N^{-1} R)^{-1}$, respectively [e.g. 1]. In this case, the Fisher matrix $M = R^\dagger N^{-1} R$ is independent of ζ . The approximate posterior uncertainty covariance as given by Eq. 55 equals the exact posterior covariance, $\Theta = (\mathbb{1} + M)^{-1} = (\mathbb{1}^{-1} + R^\dagger N^{-1} R)^{-1} = D$.

This should motivate, why this approximation can hold sensible results in sufficiently well behaved cases, in particular when a linearization of the inference problem around a reference solution (e.g. a MAP estimate) is a good approximation. Furthermore, for all signal space directions around this reference point that are unconstrained by the data, this covariance approximation returns the prior uncertainty, as it should. Additional discussion of this approximation can be found in [30], where also its performance w.r.t. ADVI is numerically investigated.

The important point about this approximate uncertainty covariance Θ is that it is a function of the latent space mean estimate θ , *i.e.* $\Theta(\theta)$, and therefore does not need to be inferred as well. For many likelihoods, the Fisher matrix is available analytically, alleviating the need to store Θ in a computer memory as an explicit matrix. It is only necessary that certain operations can be performed with Θ , like applying it to a vector or drawing samples from a Gaussian with this covariance. Relying solely on those memory inexpensive operations the MGVI algorithm is able to minimize the relevant VI KL, namely $\text{KL}_\zeta((\theta, \Theta(\theta)), d) = \mathcal{D}_{\text{KL}}(\mathcal{Q}, \mathcal{P})$, with respect to the approximate posterior mean θ . The result of MGVI are then the posterior mean θ , the uncertainty covariance $\Theta(\theta)$, and posterior samples $\{\zeta_i\}_i$ drawn according to this mean and covariance. These samples can then be propagated into posterior signal samples $s_i = s(\zeta_i)$, from which any desired posterior signal statistics can be calculated.

MGVI has enabled field inference for problems, which are too complex to be solved by MAP, in particular when multiple layers of hyperpriors were involved [e.g. 13, 15].

MGVI’s weak point, however, can be the Gaussian approximation of the posterior, which might be strongly non-Gaussian in certain applications. In order to overcome this, the *geometrical variational inference* (geoVI) algorithm [32] was introduced as an extension of MGVI. geoVI puts another coordinate transformation on top of the one used by MGVI, so that $\xi = g(y)$ approximately standardizes the posterior, $\mathcal{P}(y|d) \approx \mathcal{G}(y, \mathbb{1})$. Astonishingly, this transformation can be constructed without the (prohibitive) usage of any explicit matrix or higher order tensor in the latent space, thus allowing also to tackle very high dimensional inference problems, like MGVI. With geoVI, deeper hierarchical models, which more often exhibit non-Gaussian posteriors due to a larger number of degenerate parameters in them, can be approached via VI. Further details on geoVI and detailed comparisons of MGVI, geoVI, and Hamiltonian Monte Carlo methods can be found in [32].

5 Conclusion and Outlook

This paper argues that IFT can be regarded as a specific form of AI by summarizing recent works [29–32] that suggest this.

First, the generative models build and used in IFT are GNNs that can interpret data without initial training, thanks to the domain knowledge coded into their architecture [29]. Related architectures have very recently been proposed as image priors in AI as well [44].

Second, the VI algorithms used in IFT and AI to approximately infer quantities are a natural interface between these areas. Here, the related ADVI [47], MGVI [30], and geoVI [32] algorithms were briefly discussed, which can be used in classical AI as well as in IFT applications.

And third, the common probabilistic formulation of IFT models and GNNs, as well as the common VI infrastructure of the two areas allows for combining pre-trained GNNs and other networks with IFT style model components. In that respect, the possibility to perform Bayesian reasoning with trained neural networks as described in [31] might give an outlook into the common future of IFT and other AI methods.

In short, IFT is AI.

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